

Pancreatic Cancer Survival Analysis

Replication workflow on TCGA with a separate larger public-cohort extension

UE24CS352A · Machine Learning Mini-Project · Team members: _____

Problem

The reference project studies survival prognosis from pancreatic cancer clinical and radiomic measurements using Kaplan-Meier estimation, Cox proportional hazards, and a binary risk classifier. We reproduced the analysis structure and added Logistic Regression, RBF SVM, Random Forest, and Gradient Boosting comparisons. The Stanford cohort in the reference report is restricted; therefore, this implementation uses an open TCGA Pan-Cancer Clinical Data Resource PAAD cohort. Results below describe TCGA PAAD and are not a numerical replication of Jamalain's cohort.

Dataset

The NCI GDC TCGA-CDR table yielded 184 PAAD records with overall-survival time and status (100 observed deaths; 84 right-censored observations). Clinical predictors used for classification were age, gender, tumor stage, histology, and race where available. The Cox model used age, gender, and stage after one-hot encoding and fold-local missing-value imputation. No radiomic features are present in this source. For binary classification, the target is observed death by 270 days; the 13 patients censored before 270 days were omitted because their status at that horizon is unknown. This left 171 known labels, including 32 high-risk events.

Approach

We evaluated a penalized Cox proportional-hazards model using five-fold shuffled cross-validation and Harrell's C-index. The separate binary task used stratified five-fold cross-validation with preprocessing fitted inside each training fold. Naive Bayes used quantile-discretized numerical features; other classifiers used imputation and scaling/one-hot encoding as appropriate. We report balanced accuracy, precision, recall, F1, and ROC-AUC. The high-risk horizon was also compared at 240, 300, 330, and 360 days. We did not apply SMOTE: interpolating right-censored times can create invalid survival observations, and oversampling before cross-validation can leak information.

What differs from the source report

The original report uses a Naive Bayes model to select among survival thresholds, generates Kaplan-Meier curves for model-defined risk groups, and compares clinical, radiomic, and combined Cox models. This project reproduces the survival/C-index and risk-classification structure where TCGA-CDR supports it, adds extra classifiers, and reports a descriptive Kaplan-Meier/log-rank analysis. It cannot evaluate radiomic or combined feature sets because the public extract has clinical variables only.

Results and Interpretation

Five-fold evaluation on the TCGA PAAD substitute cohort

Binary classifier comparison at 270 days

Classifier	Bal. acc.	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.642	0.282	0.688	0.400	0.620
SVM (RBF)	0.640	0.274	0.719	0.397	0.417
Random Forest	0.608	0.300	0.469	0.366	0.574
Naive Bayes	0.574	0.235	0.594	0.336	0.602
Gradient Boosting	0.518	0.273	0.094	0.140	0.583

Fixed Logistic Regression and RBF SVM led TCGA (balanced accuracy 0.642 and 0.640). Low precision reflects the 32 early deaths; discrimination remained modest.

Larger public-cohort extension and nested tuning

On cBioPortal MSK (2027 known labels), fixed Logistic Regression reached 0.653 balanced accuracy / 0.692 ROC-AUC; the 3-feature Cox C-index was 0.657 (SD 0.016). Nested 5-fold outer / 3-fold inner tuning scored SVM 0.661 (fixed 0.652) and Random Forest 0.658 (fixed 0.590); tuned Logistic Regression was 0.650 (fixed 0.653). On TCGA, tuned Naive Bayes rose from 0.574 to 0.606, while tuned Logistic Regression was 0.624 (fixed 0.642). These are cohort-specific estimates, not like-for-like gains over Jamalian.

Naive Bayes threshold sweep

Horizon	Known labels	High risk	Balanced accuracy
240 days	173	29	0.543
270 days	171	32	0.574
300 days	167	37	0.493
330 days	165	40	0.515
360 days	162	42	0.512

TCGA Cox mean C-index was 0.522 (SD 0.035; 5/5 folds). Its full-cohort median-risk log-rank $p=0.0012$ is descriptive, not validation. Jamalian reports 0.67-0.82, but cohort and predictors differ.

Conclusion and limitations

Nested tuning helped some models and hurt others. The MSK extension gives more stable cohort-level estimates, but neither dataset supports clinical use. Report the cohort, censoring exclusions, feature set, and nested evaluation whenever quoting these scores.

Sources: Jamalian, A. (2020), CS229 Project Final Report: Pancreatic cancer prognosis using clinical and radiomic data. TCGA-CDR / NCI Genomic Data Commons, <https://gdc.cancer.gov/about-data/publications/PanCan-Clinical-2018> . MSK cohort: cBioPortal PDAC MSK 2024, https://www.cbioportal.org/study/summary?id=pdac_msk_2024 . Code, seeds, and per-model metrics are in the accompanying repository outputs.